Time Series - 5

Agenda ⇒

Expo smoothing
ARIMA
SARIMAX
exog
Fb
Seasonality

endog V/s exog

Holidays
Weekends
Diwali

XGB GBDT, Linear Reg

→ Time Series as Regression

PROPHET FB

Use cases

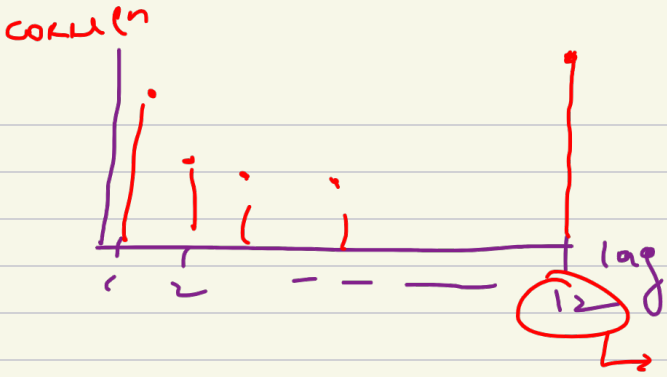
Desktop Browser

Weekend Holiday

Sales lag Sales

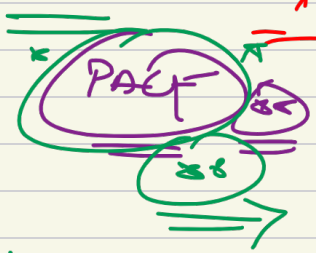
t
t-1
t-2

ACF & PACF



Seasonality = 12

Restaurant visits



Total no: of visitors

holidays → exog

exog → should be available for future date as well

Valid exog → Training → Testing

Holiday → Historical + future

features → valid ?

DATA

SARIMAX → X

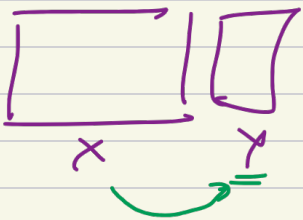
Calendar →

DATE → sales

TS as Regression ⇒

visitors | sales → single feature

gBDT, RF, pTree

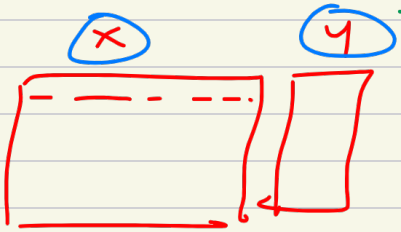


i/p → o/p

.fit()

X & Y

$y = f(x)$



gBDT, RF, ...

formula

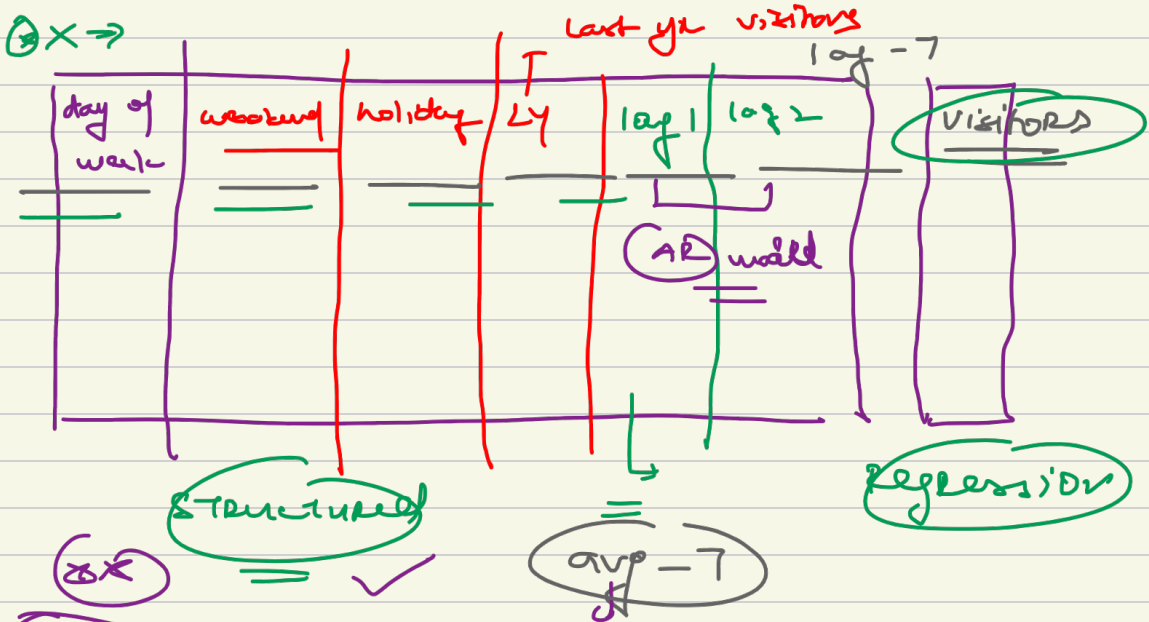
→ linear Reg~

ML

80% → feature creation, Preprocessing, Engineering

DATA Sales

model + SKlearn + HP Tuning



Linear, DTREE, RF ⇒ evaluate MAPE

FEATURE engineering

Better features ⇒ Better performance

Quiz time!

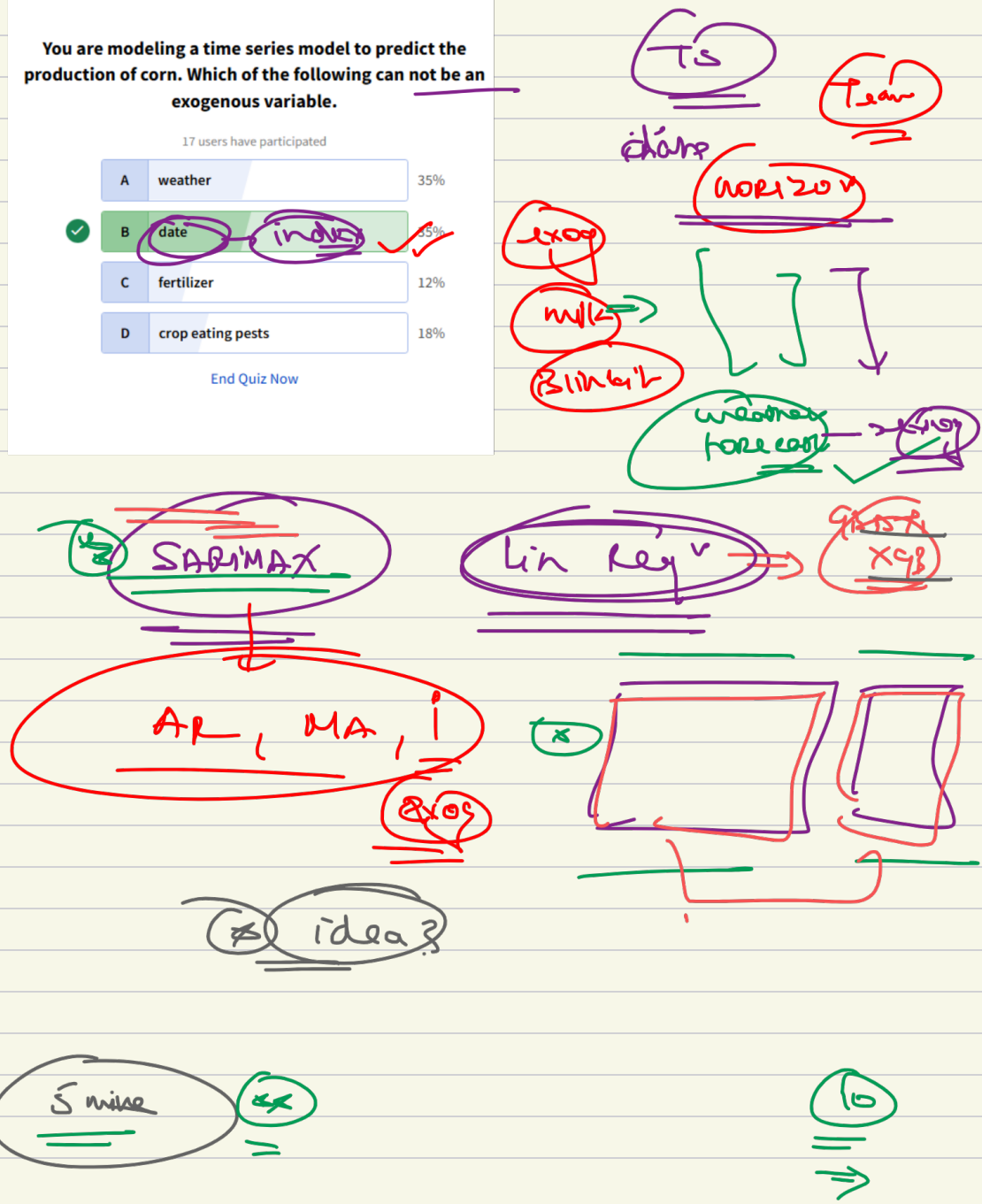
Time Left: 28s

You are modeling a time series model to predict the production of corn. Which of the following can not be an exogenous variable.

17 users have participated

A	weather	35%
B	date	35%
C	fertilizer	12%
D	crop eating pests	18%

End Quiz Now



FB PROPHET

LIBRARY

FB ⇒ 2017

→ supports multiple seasonality

hour | week | year

From company

DATE	ORDERS	
Jan 1	100	Historical
2	104	
.	107	
.		
.		
31	180	

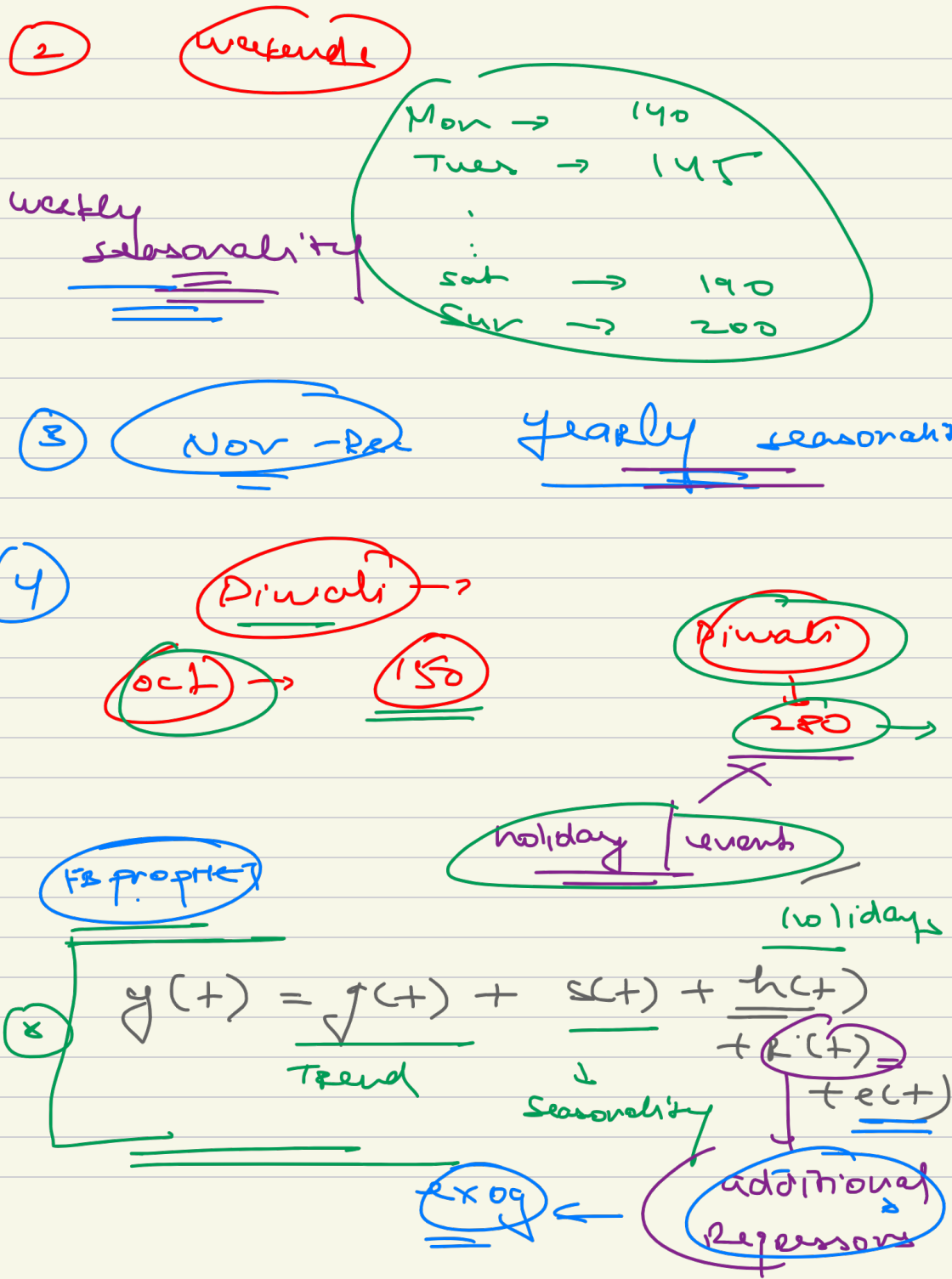
Every day for next 3 months

→ 1

Year ago → 100 orders / day

Recent → 160 order / day

Trend → −


2
weekends
Mon → 140
Tues → 145
Sat → 190
Sun → 200
weekly seasonality
3
Nov - Dec
yearly seasonality
4
Diwali
Oct → 150
Diwali
250
holiday events
FB prophet
holidays
x
y(t) = g(t) + s(t) + h(t) + ε(t)
Trend
Seasonality
exog
additional Regressors

Prophet

holiday

Sunday (Nov 6)

Prophet

Trend = 160

Sunday = +25

Nov = +10

Diwali = +45

— (Marketing = +20

FB prophet

Forecast = 260

Prophet

lower

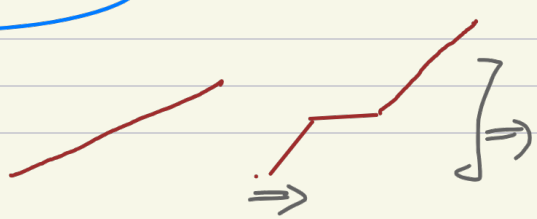
t-1 t-2

AR, MA

ARIMA

Prophet

Trend



Change Points

Prophet + Change points

2000 / month

After July → 8000 / month

mobile app

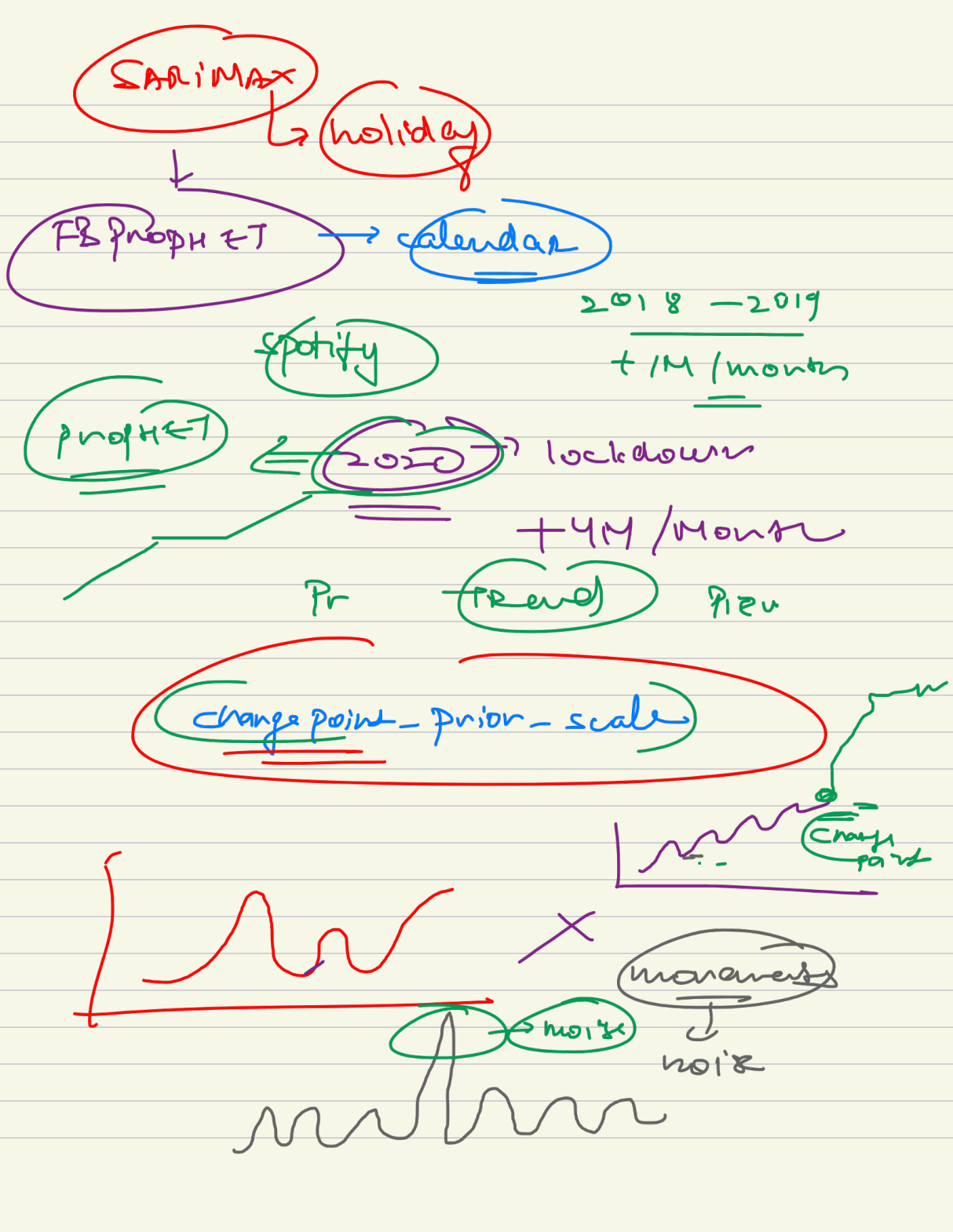
July → ChangePoint

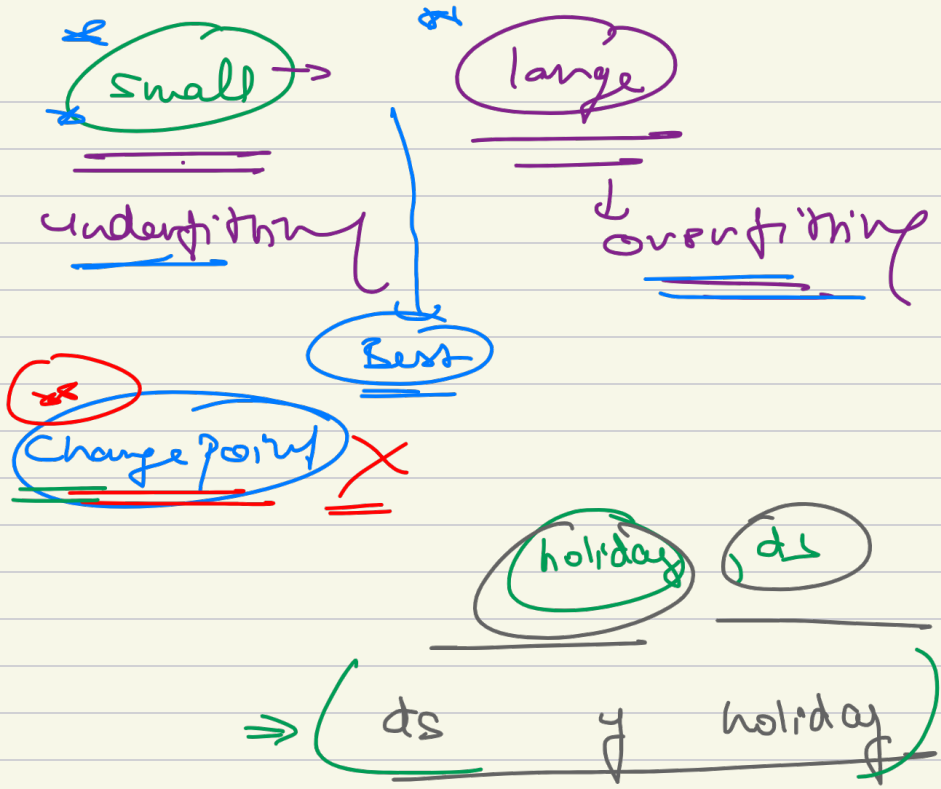
Before +2k / month

After → +8k / month

Prophet

Piecewise Trend


SARIMAX
holiday
FB PROPHET
calendar
2018 – 2019
+1M / month
spotify
prophet
2020
lockdown
+4M / month
Pr
trend
Piev
change point_prior_scale
change point
noise
noise


small
large
underfitting
overfitting
Best
Change point
holiday
ds
ds
y
holiday

TS problem

Gold
PRIMA
Project

Beginner

Series

Steps

Business que

Forecast sales × Profit

Units sold

granularity → STORE × SKU

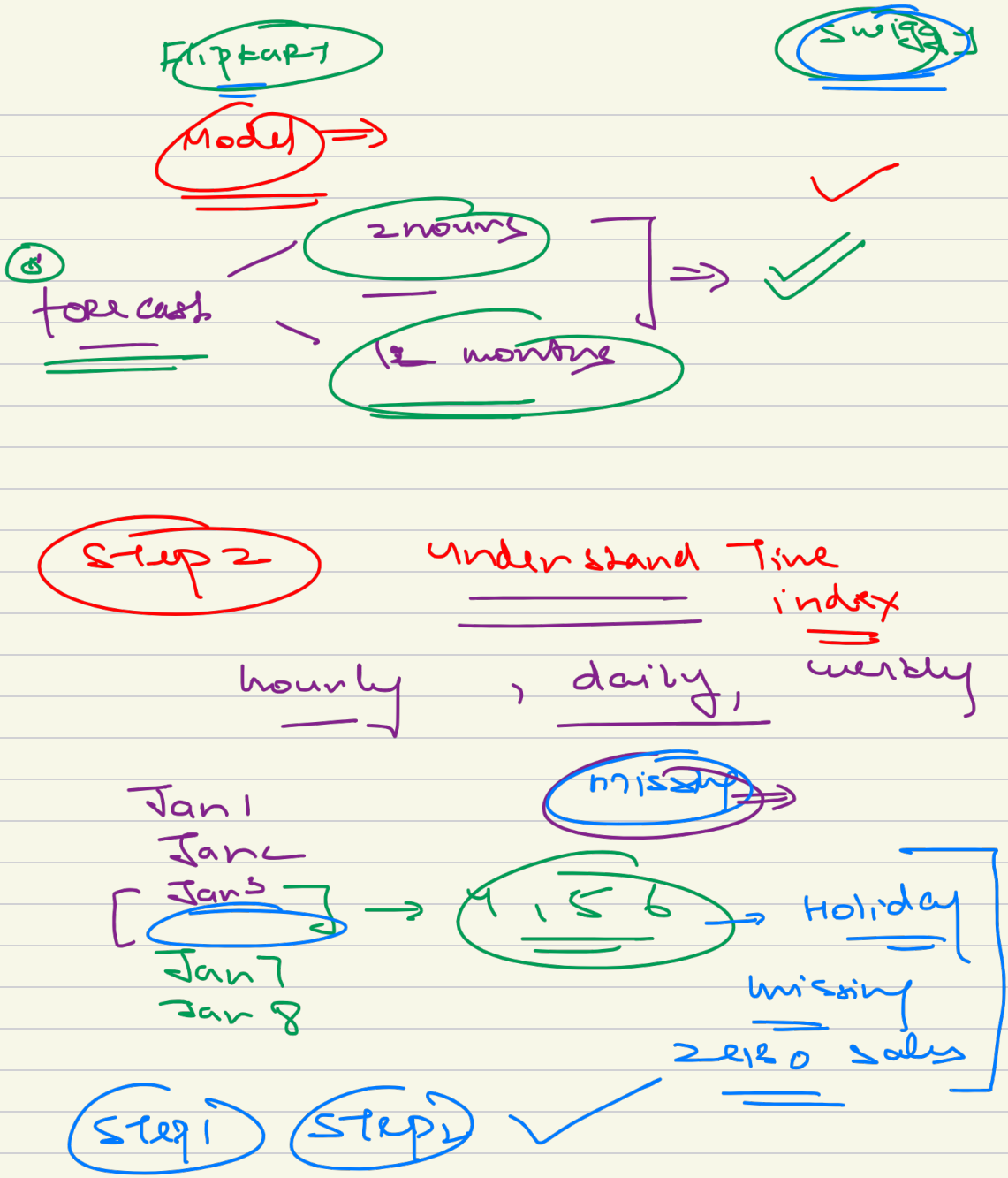
Freq → Daily

Horizon → Next 28 days

why → inventory

Cost of underforecasting → Stockouts

Overforecasting → Excess inventory


Flipkart
Swiggy
Model
2 hours
forecast
& monitor
Step 2
understand Time index
hourly , daily, weekly
Jan 1
Jan 2
Jan 3
Jan 7
Jan 8
missing
4, 5, 6
Holiday
missing
zero sales
Step 1
Step 2

Step 3 ⇒ Plot TS

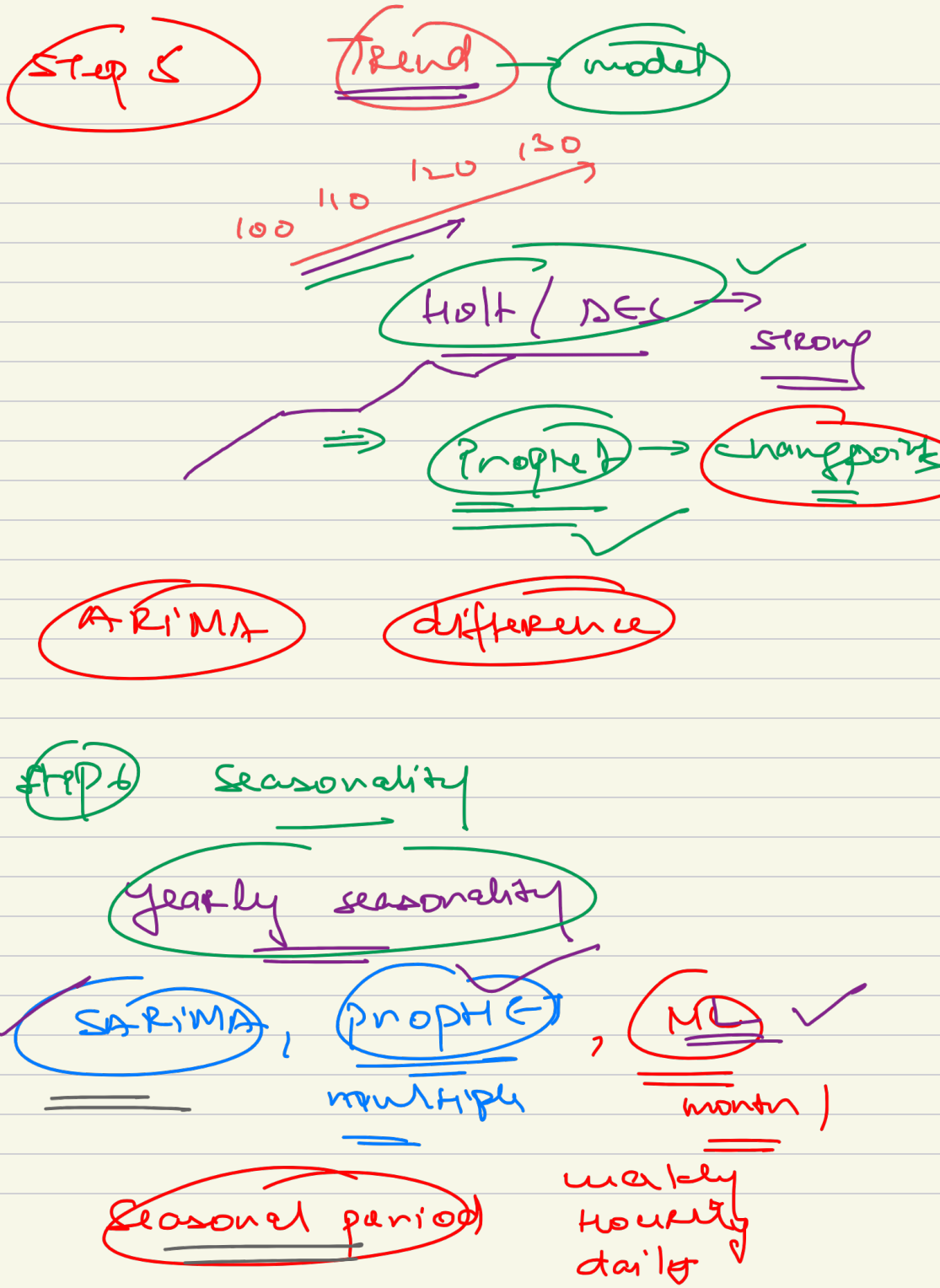
eye

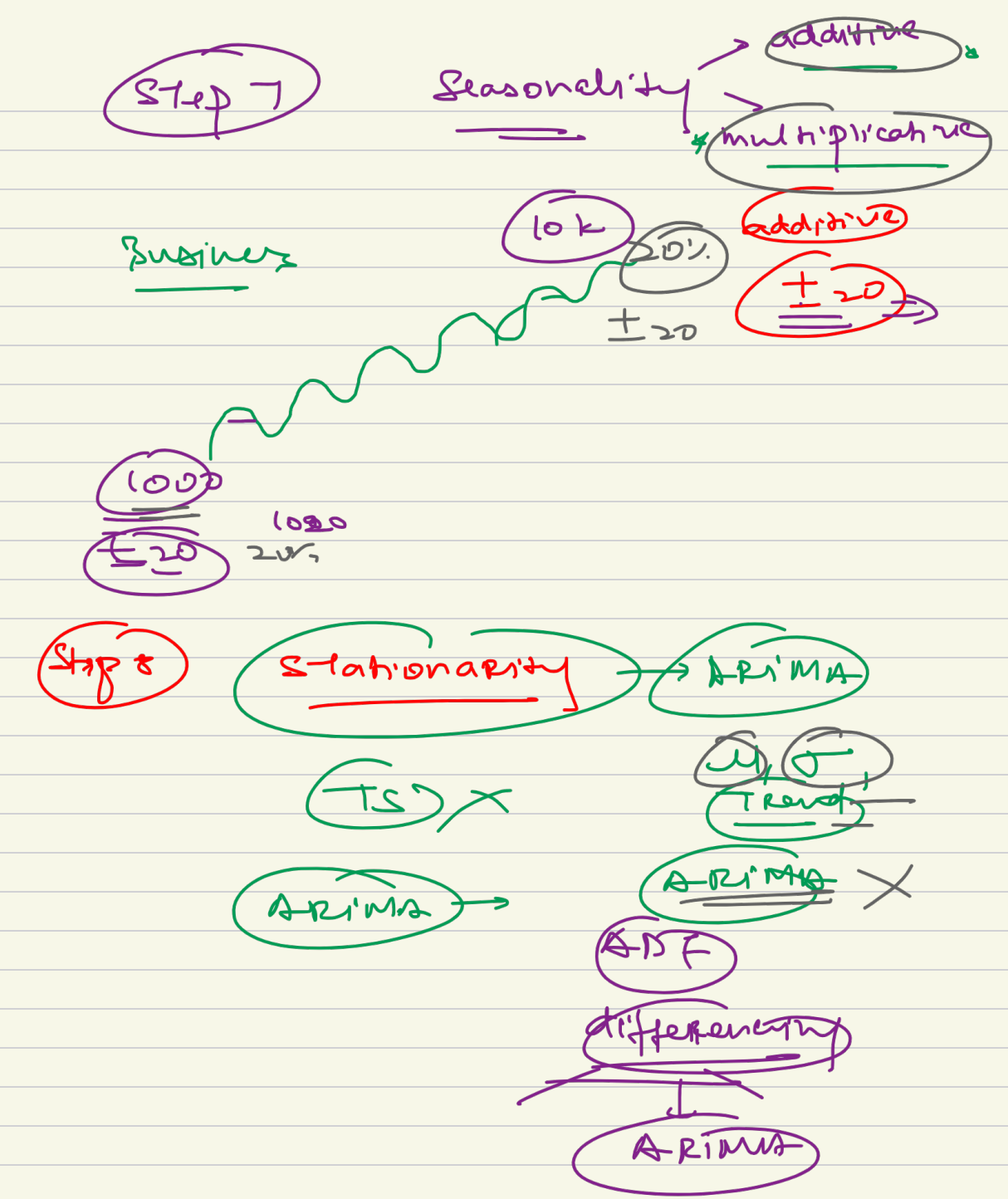
Seasonality ⇒ Stationary

Trend ↑ ↓

Step 4 Search for patterns

1. Trend → ↑ or ↓
2. Seasonality →
3. Multiple seasonality
4. Changing variance → change point
5. ACF ⇒ ✓
6. Outliers →
7. External drivers → Promotions


Step 5
Trend
model
100
110
120
130
Holt / SES
strong
Prophet
changepoints
ARIMA
difference
Step 6
Seasonality
yearly seasonality
SARIMA
,
prophet
,
ML
multiply
month
Seasonal period
weekly
hourly
daily


Step 7
Seasonality
additive
multiplicative
Business
10k
20%
±20
additive
±20
1000
±20
1020
2%
Step 8
Stationarity
ARIMA
TS
ARIMA
M, σ
Trend
ARIMA
ADF
differencing
ARIMA

Exterval ⟹ SARIMAX

Icecream sales ⟹

temp, Rain, holiday, promotion

Sales$_t$ = Time series + weather + Promotion

SARIMAX

PROPHET ✓

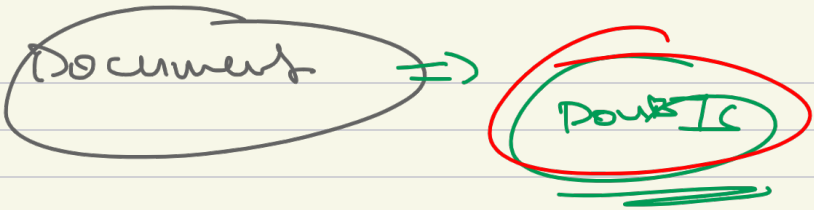
ML ⟹

Baseline model

SARIMA(2,1,2)(1,1,1,12)

Naive ⇒ $\hat{y}_{t+1} = y_t$


Seasonal Naive
Naive
Prophet
SARIMA
8%.
7.8%.
7.9%.
Model
evaluating
Residuals
residuals
weekly
yearly
ideas


Document
DocID